

Sensitivity of Antarctic sea level contribution to bed friction inversion choices in the BISICLES ice sheet model

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Abstract. Antarctica is the largest potential contributor to future sea level but is also the most uncertain. The ice sheet initial state is known to be a significant source of uncertainty, but this uncertainty is rarely explored outside of large-scale model intercomparison projects.

Century-scale projections of the Antarctic ice sheet often generate an initial state by solving an inverse problem for basal friction — and sometimes other fields — such that modelled ice velocities closely match observations. While this yields realistic short-term behaviour, it may also overfit by masking deficiencies elsewhere in the model.

Few modelling studies explicitly treat the inverse problem itself as a source of uncertainty. There is some evidence that regularisation choices within the inversion can produce uncertainty comparable to other commonly perturbed ice sheet model parameters, though this is limited to select experimental setups. It is unknown whether these results translate to other ice sheet models or domains.

Here, we use the BISICLES ice sheet model to explore inverse problems as a source of uncertainty in Antarctica's future sea level contribution. We generate an ensemble of initial states and project them forward to 2300 under a high-emission scenario. We assess the sensitivity of sea level projections to regularisation parameters within the inversion and compare it against other commonly perturbed parameters. We find that the sensitivity to regularisation in BISICLES can be high on century timescales, but that this declines for more long-term projections.

1 Introduction

Note: do I need a more broad introduction to Antarctica here? So many papers give an overview of recent sea level change, big picture dynamics, MISI, etc. It doesn't feel as necessary for Cryosphere, but maybe it is.

The Antarctic ice sheet is a major potential source of future sea level rise, but its projected sea level contribution is highly uncertain (Seroussi et al., 2020; Edwards et al., 2021). In particular, the upper ranges of this uncertainty are important for climate adaptation strategies such as sea defenses, and underestimating this uncertainty poses a risk to the resilience of coastal communities (citation needed?). Modelling studies that evaluate this uncertainty should therefore seek to include as many relevant types of uncertainty as their experiment will allow. Most commonly, this will include uncertainty in model parameters or in the climate forcing (Bulthuis et al., 2019; Hill et al., 2021; Coulon et al., 2024, 2025) (are these enough / the most

25 appropriate refs? There are so many possible!). Some studies examine structural uncertainty in ice sheet model components such as basal sliding or sub-shelf melt parameterisations (Barnes and Gudmundsson, 2022; Burgard et al., 2022) (These are maybe the best examples of purely structural uncertainty explorationm but lots of possible citations that explore structural a little bit alongside a more parametric uncertainty focus). in comparison, uncertainty in the ice sheet initial state is less well-explored, despite being a large source of uncertainty in studies that include it (Seroussi et al., 2019; Rosier et al., 2025).

30 The initial state refers to a set of spatial fields defining the physical condition of the ice sheet, which can include ice thickness, bedrock elevation, ice temperature or enthalpy, basal properties, damage, and others. Ideally, these would be similar to the modern ice sheet whilst staying dynamically consistent with observations. However, this is not straightforward to achieve, mainly due to inconsistencies between datasets, the model physics, and observed dynamics (citation needed?). Some models overcome this by using inverse methods to solve for certain components of the initial state, such as a basal friction coefficient
35 (Larour et al., 2012; Cornford et al., 2015). Inverse methods can produce initial states with very good fits to observed dynamics, but they also mask deficiencies in the model and therefore underrepresent uncertainty. This is not such an issue for short-term projections where the modelled ice sheet stays close to its present-day state, but it may be more problematic for longer-term projections with large changes in ice geometry and dynamics. Furthermore, inverse problems are underdetermined, since they are often solving for more parameters than they have data, meaning that no unique solution exists. Other equally plausible
40 solutions to the inverse problem could yield different future sea level outcomes, but these are rarely explored.

Standard approaches to the inverse problem seek to minimise an objective function that includes a term for model-observation misfit. However, the minimum of this function contains no information about the uncertainty in the solution. For this, one would need to know the curvature of the function around the minimum, which is described by the inverse of the Hessian matrix. However, for practical ice sheet modelling purposes, the Hessian is too large to compute directly. Several methods have been
45 developed to approximate this process, but these are still computationally expensive and have therefore only been applied to highly idealised problems (cite). Until these methods become more tractable, it is not clear how best to explore uncertainty in the initial state in a way that is physically plausible. The simplest idea is to scale the inverted fields by a scalar multiplier (Nias et al., 2016; Bevan et al., 2023), but this only allows for spatially homogeneous perturbations and does not explore uncertainty in the spatial distribution of parameters. Several studies have perturbed the bedrock elevation and solved inverse problems
50 for each realisation (cite), but great care must be taken to properly account for spatially-varying autocorrelated errors in the bedrock data (cite andreas). A recent third approach is to treat hyperparameters within the inversion as uncertain parameters and produce ensembles of initial states by perturbing these (Rosier et al., 2025). This allows for spatially-varying perturbations that are physically plausible by design. It also converts initial state uncertainty into more conventional parametric uncertainty, to which sensitivity analysis methods can be applied.

55 Methods to solve the inverse problem vary between models, but most rely on some form of regularisation. If unregularised, the optimisation tends to introduce short-wavelength, high-amplitude variability into the inferred fields, even with only a minor benefit to the observation misfit. Regularisation imposes structure on the solution by penalising roughness and encouraging smooth solutions. However, it is not clear what is the appropriate amount to regularise a solution. An L-curve heuristic is commonly used to justify a choice of regularisation strength by finding the optimal compromise between proposed ‘over’ and

60 ‘under’-fitting, characterised by the corner of the ‘L’ (Hansen, 1992). This does not guarantee, however, that the true solution is one that minimises the roughness, nor that a good fit is truly representative of real basal properties. Furthermore, Rosier et al. (2025) demonstrate that projected sea level contribution can be highly sensitive to regularisation in the inversion, along with other initialisation hyperparameters. This suggests that regularisation hyperparameters may be an important route to exploring initial state uncertainty in sea level projections, but it is not clear whether this result holds outside of the specific model or
65 domain.

To further study uncertainty in the ice sheet initial state, we use the BISICLES ice sheet model to assess the sensitivity of whole-Antarctic sea level contribution to regularisation parameters in the inverse problem. We produce an ensemble of initial states by perturbing these parameters, alongside other commonly perturbed ice sheet model parameters, and project them forward to 2300 under a high-emission scenario. We find that regularisation parameters can be a moderate source of
70 uncertainty in century-scale sea level projections, but that this sensitivity is likely not consistent across models, domains, and timescales. We discuss the implications of these results for future research efforts to explore initial state uncertainty in sea level projections.

2 Methods

2.1 BISICLES ice sheet model

75 BISICLES (Cornford et al., 2013) is an adaptive mesh, vertically integrated ice sheet model, which has been used extensively for century-scale modelling of the Antarctic ice sheet (Cornford et al., 2015; Siahaan et al., 2022; O’Neill et al., 2025). It uses the L1L2 approximation to Stokes flow (Schoof and Hindmarsh, 2010), which incorporates membrane stresses with a simplified treatment of vertical shear, allowing it to simulate a wide range of flow regimes in Antarctica with computational efficiency. Here, we run BISICLES with a base resolution of 8 km and 3 layers of refinement, up to 1 km at the grounding
80 line and in regions of fast-flowing ice. This resolution significantly improves the representation of grounding line dynamics compared to coarser grids (Cornford et al., 2016).

BISICLES solves the 2-dimensional L1L2 mass transport and stress balance equations outlined in Cornford et al. (2015). They feature a vertically integrated effective viscosity, $\phi h \bar{\mu}$, defined by:

$$\phi h \bar{\mu}(x, y) = \phi \int_{s-h}^s \mu(x, y, z) dz, \quad (1)$$

85 where h is ice thickness, s is surface elevation, and ϕ is an uncertain, spatially-varying coefficient that is determined by the inverse problem. ϕ can be interpreted as a damage multiplier and typically takes values between 0.05 and 1, with lower values in regions close to the grounding line or in fast-flowing ice streams. The vertically varying effective viscosity, $\mu(x, y, z)$, is calculated using the Glen-Steinemann flow law (Glen, 1955):

$$\mu(x, y, z) = \frac{1}{2A_0 A(T) \tau^{n-1}}, \quad (2)$$

90 where $A(T)$ is a temperature-dependant Paterson rate factor (Cuffey and Paterson, 2010). The 3-dimensional temperature field is derived from a 100,000-year thermal spin-up described in Trevers et al. (2024). τ is the deviatoric stress tensor and the flow law exponent n is an uncertain parameter that we perturb in the ensemble. A correction factor, $A_0 = 100,000^{(3-n)}$, ensures that the effective viscosity is independant of n at a reference stress of 100 kPa. Either side of this value, the flow law diverges from the $n = 3$ case, but the effective viscosity is scaled by ϕ during the inverse problem so that the initial state matches observed ice velocities. Antarctic sea level projections have been shown to be very sensitive to the choice of n in BISICLES and in other ice sheet models (Martin et al., 2026; Getraer and Morlighem, 2025).

Basal sliding is governed by a pressure-limited Weertman-Coulomb sliding law derived from Tsai et al. (2015), which allows for Weertman and Coulomb-like behaviour over hard and soft beds, respectively. The basal shear stress, τ_b , is given by:

$$\tau_b = -\min(\tau_w, \tau_c) \frac{\mathbf{u}}{|\mathbf{u}|}, \quad (3)$$

100 where τ_w has a power-law relationship with the velocity $\mathbf{u}$, and τ_c is a Coulomb-limiting term defined by the effective pressure, N :

$$\tau_w = C|\mathbf{u}|^m, \quad (4)$$

$$\tau_c = \frac{1}{2}N. \quad (5)$$

C is a spatially varying friction coefficient that, along with the viscosity coefficient, ϕ , is determined via the inverse problem.

105 We treat the friction exponent m as an uncertain parameter which is to be perturbed in the ensemble. This sliding law has been used in previous BISICLES experiments (O'Neill et al., 2025) and performs similarly to other common sliding laws in Antarctica (Barnes and Gudmundsson, 2022).

2.2 Model initialisation

We initialise BISICLES using an inverse method to solve for the basal friction coefficient, C , and the viscosity coefficient, ϕ , such that modelled surface velocities closely match observations. BISICLES uses conjugate gradient descent (Shewchuk, 1994) to minimise a cost function, J , defined by:

$$J = J_m + J_p, \quad (6)$$

where J_m is a function of the velocity misfit

$$J_m = \frac{1}{2} \int_{\Omega_V} (|\mathbf{u}| - |\mathbf{u}_{obs}|)^2 d\Omega_V, \quad (7)$$

115 and J_p is a Tikhonov penalty function

$$J_p = \frac{\alpha_C}{2} \int_{\Omega_V} |\nabla C|^2 d\Omega_V + \frac{\alpha_\phi}{2} \int_{\Omega_V} |\nabla \phi|^2 d\Omega_V. \quad (8)$$

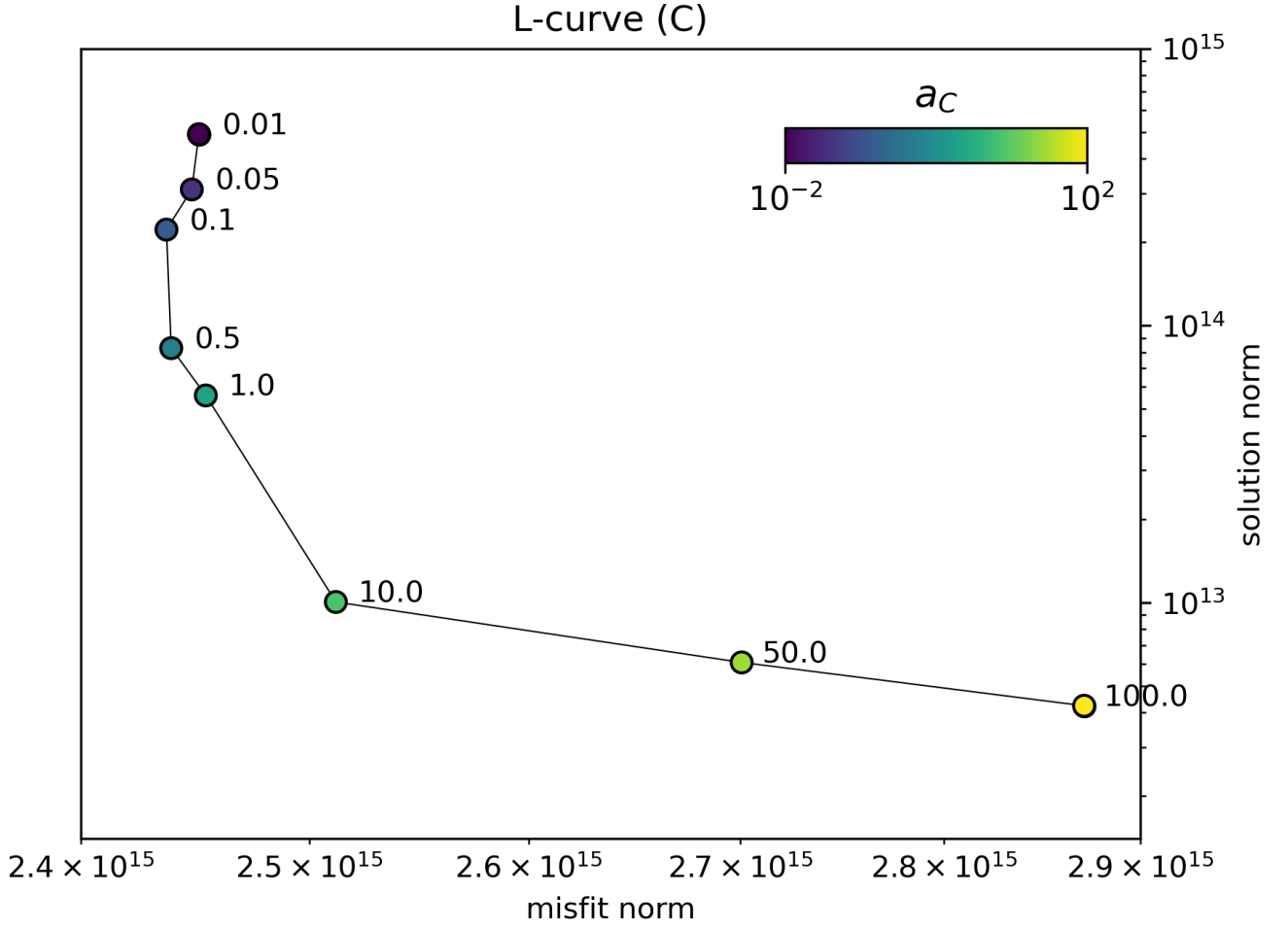


Figure 1. L-curves

α_C and α_ϕ are regularisation parameters that control the smoothness of C and ϕ , respectively. In this study, we treat both as uncertain parameters within a range of values determined by L-curve analysis. Ω_V is the domain over which velocities are observed, and $\mathbf{u}_{obs}$ are the observed surface velocities from Rignot et al. (2017).

120 Since the Tsai sliding law is independent of C in areas of low hydrostatic pressure, we solve the inverse problem using a simple linear sliding law that constrains C everywhere. For ensemble members with $m \neq 1$, we then adjust C post-inversion such that basal drag (and therefore ice velocity) is equal on either side of the initialisation. This is done by scaling C according to

$$C_m = C_{(m=1)}(|\mathbf{u}| + \epsilon)^{1-m}, \quad (9)$$

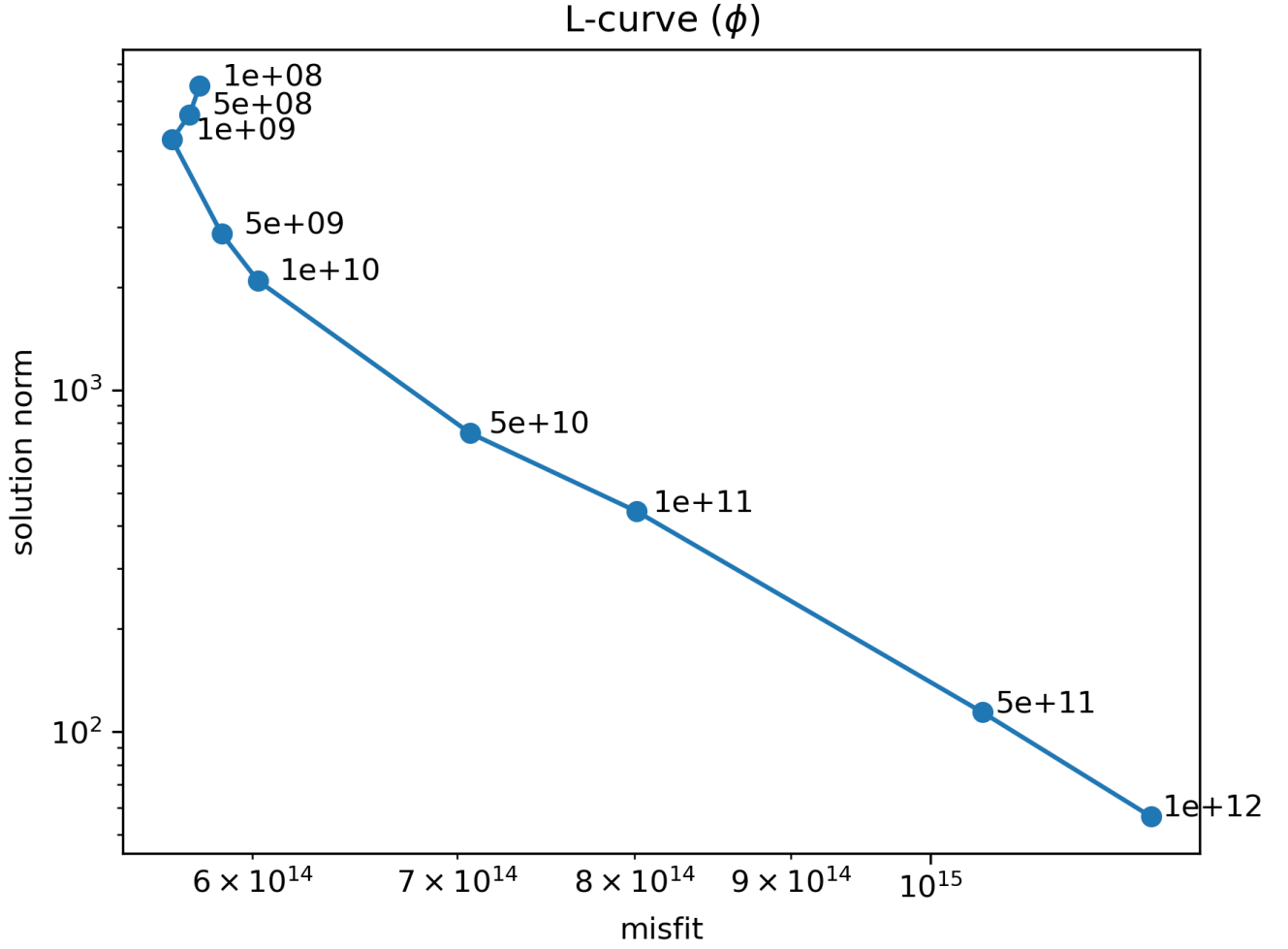


Figure 2. L-curves

125 where $\epsilon = 10^{-5}$ is a small regularising term to prevent infinite friction values. Ideally, this would yield the same solution for basal friction regardless of the value of m used in the inversion, although there can be differences due to the path-dependence of the solution (Wolovick et al., 2023). Nonetheless, we found that inversions with a linear sliding law were faster and more convenient than with $m \neq 1$.

The initial ice geometry is taken from BedMachine Antarctica v2 (Morlighem, 2020), the initial guess for ϕ is uniformly 1, 130 and the initial guess for C is calculated by equating basal drag to the driving stress. This yields

$$C_{init} = \rho_i g h \nabla s |\mathbf{u}|^{-1}, \quad (10)$$

where $\rho_i = 917 \text{ kg m}^{-3}$ is the density of ice and $g = 9.81$ is gravitational acceleration.

As is common with initialisation by inverse problem, we observe a short-term shock immediately after initialisation as the model adjusts to inconsistencies between the geometry and model physics. We therefore allow a 15-year relaxation period, during which we periodically resolve the inverse problem following Bevan et al. (2023). Throughout the relaxation, the thickness of floating ice shelves is held constant and we apply a surface mass balance from the MAR regional climate model (Agosta et al., 2019).

2.3 Climate forcing

Inverse methods produce a good fit to observed velocities, but this often results in poor fits to other measurables such as the rate of elevation change, $\dot{h}$. We could incorporate an additional term into the cost function (7) to improve the fit to $\dot{h}$, but this would come at a cost to the velocity misfit. Model outputs can also be very sensitive to the balance between the velocity and $\dot{h}$ components of a misfit function (Rosier et al., 2025). Instead, we choose to correct for bias in $\dot{h}$ by exploiting uncertainty in the surface mass balance SMB. We apply a background rate of surface mass balance that is designed to bridge the gap between modelled and observed $\dot{h}$:

$$\text{SMB}_{bkg} = \dot{h}_{obs} + \nabla \cdot (\mathbf{u}h) \quad (11)$$

where $\dot{h}_{obs}$ is the observed rate of elevation change from (Smith et al., 2020) and $\nabla \cdot (\mathbf{u}h)$ is the flux divergence at the end of a relaxation with default parameter values. We set bounds of -1 – 2 ma^{-1} and apply a Gaussian filter to smooth out short-wavelength components of the flux divergence (Nias et al., 2016). The beneficial effect of this is to correct a thickening bias over the Antarctic Peninsula and trans-Antarctic mountains, where our 1 km grid resolution is too coarse to resolve the dynamics of these smaller glaciers. However, the resulting surface mass balance is purely synthetic and a poor representation of reality. Nonetheless, it corrects the significant model drift that occurs over multiple centuries without the need for a control simulation. We therefore consider it to be adequate for the purposes of a sensitivity study.

On top of this background surface mass balance, we apply a time-varying anomaly from the CESM2-WACCM climate model (Danabasoglu et al., 2020) under the SSP5–8.5 emissions scenario, calculated following the ISMIP6 protocol (Nowicki et al., 2020). This was chosen for its strong climate forcing to maximise the signal-to-noise ratio in our results, but is not necessarily representative of the most likely future climate.

We also apply a basal mass balance to floating ice using a quadratic parameterisation of ice shelf basal melt with respect to ocean thermal forcing (Jourdain et al., 2020),

$$m(x, y) = \gamma_0 \times \left(\frac{\rho_{sw} c_{pw}}{\rho_i L_f} \right)^2 \quad (12)$$

$$\times (\text{TF}(x, y, z_{\text{draft}}) - \delta T_{\text{sector}}) \quad (13)$$

$$\times |\langle \text{TF} \rangle_{\text{draft} \in \text{sector}} + \delta T_{\text{sector}}|, \quad (14)$$

Where γ_0 is an uncertain parameter that controls the sensitivity of ice shelf melt to increases in ocean temperature and salinity. ρ_{sw} and c_{pw} are the density and specific heat capacity of seawater, ρ_i is the density of ice, and L_f is the latent heat of fusion.

Table 1. Perturbed parameters used in the ensemble.

Parameter	Long name	Min.	Nominal	Max.
α_C	C -regularisation	10^{-2}	10^0	10^2
α_ϕ	ϕ -regularisation	10^8	10^{10}	10^{12}
n	Flow law exponent	2	3	4
m	Sliding exponent	1	3	5
γ	Ice-ocean heat flux	9619	39131	159189
UMV	Upper mantle viscosity	6×10^{18}	2.45×10^{19}	10^{21}

TF(x, y, z_{draft}) is the ocean thermal forcing anomaly at the depth of the ice shelf draft, calculated from ocean temperature and salinity outputs from CESM2-WACCM. Anomalies are calculated with respect to the initial thermal forcing such that basal mass balance is zero upon initialisation. δT_{sector} is a sector-specific offset according to the ‘MeanAnt’ calibrated values in Jourdain et al. (2020).

2.4 Experiment design

We consider six parameters in BISICLES, including two regularisation parameters in the inverse problem (α_C and α_ϕ). The four remaining parameters are selected for their prevalence in other sensitivity studies and their breadth across multiple ice sheet model components (Table 1). To explore the sensitivity to ice rheology, we perturb the flow law exponent, n (Getraer and Morlighem, 2025; Martin et al., 2026); for sensitivity to the form of the basal sliding relation, we perturb the friction exponent, m (Ritz et al., 2015; Brondex et al., 2019); for sensitivity to the rate of ice shelf melt, we perturb the ice-ocean heat flux, γ (Hill et al., 2021; Coulon et al., 2024, 2025); and for sensitivity to the solid-Earth response, we perturb the upper mantle viscosity, UMV (Pollard et al., 2017; Kachuck et al., 2020).

We produce an ‘axial’ ensemble, sampling 6-dimensional parameter space by perturbing one parameter at a time to its minimum and maximum value while holding the other parameters at their nominal value. The nominal value is either the arithmetic or geometric mean of its upper and lower bounds, and this is chosen for the regularisation parameters to coincide with the corner of the ‘L’ (Figure). Along with a central simulation with no parameters perturbed, this yields a total of 13 ensemble members. For each, we initialise by solving the inverse problem and then project the initial state forward under the same climate forcing. We consider the sensitivity of Antarctic sea level contribution to each parameter by finding the difference in values between simulations with its maximal and minimal value. This is a much more rudimentary measure of sensitivity than, for example, Sobol’ sensitivity (Hill et al., 2021; Rosier et al., 2025), and it doesn’t provide any information about the shape of the relationship between parameters and outputs, as would be possible with a full perturbed parameter ensemble. However, it requires far fewer simulations and is sufficient for a preliminary exploration of sensitivity before investing in a more comprehensive experiment design.

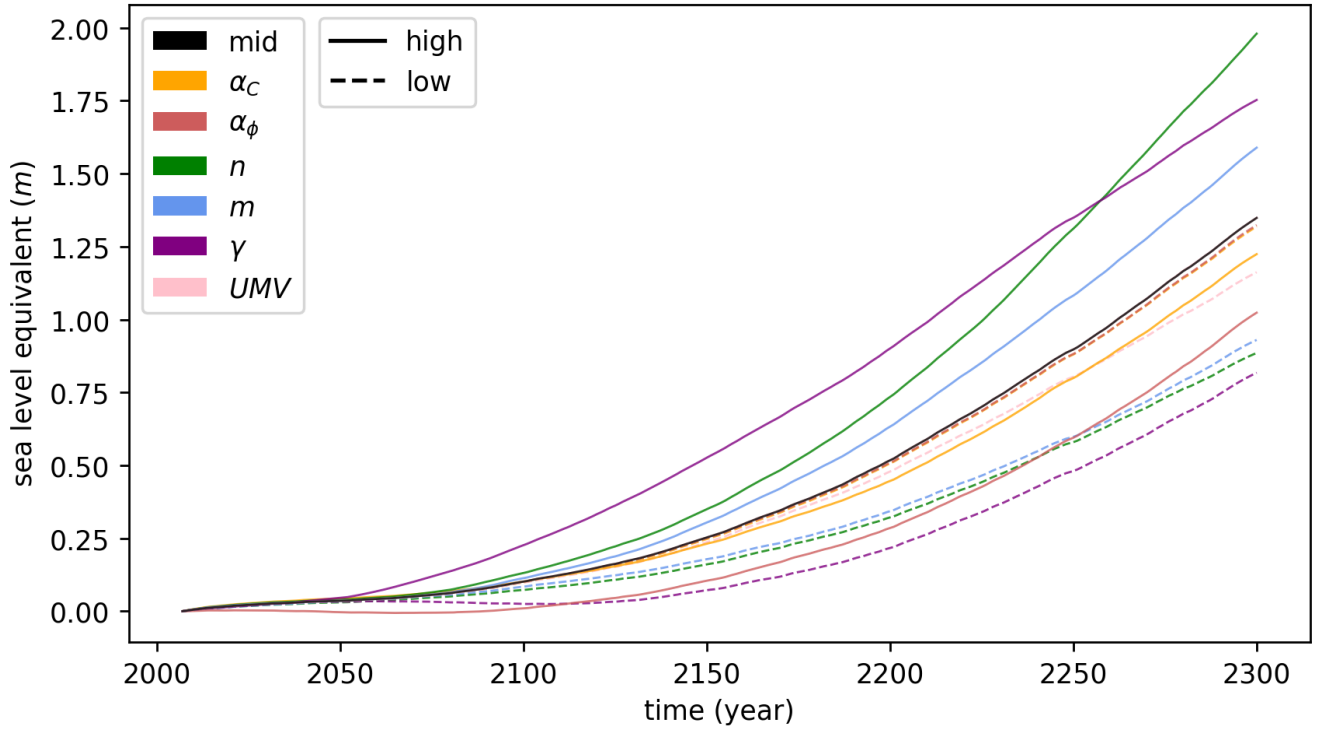


Figure 3. Timeseries of Antarctic sea level contribution for high and low values of each perturbed parameter.

3 Results

We project our initial states forward with the CESM2 climate forcing up to 2300 and calculate total Antarctic sea level contribution for each ensemble member (Figure 3). The ensemble range at 2100 is ??–?? m, which compares well with the IPCC
 190 AR6 range of 0.03–0.28 m under SSP5–8.5 (cite IPCC). In the 22nd and 23rd centuries, all ensemble members accelerate their mass loss and the range widens to ??–?? m by 2300.

The spatial patterns of ice thickness change are broadly similar across the ensemble, with most mass loss occurring in regions of West Antarctica: the Amundsen Sea sector, the Siple coast, and the Weddell sea sector. However, the extent of mass loss in these regions depends on parameter choices, with high variance along Thwaites, Foundations, and Denman glaciers (Figure 4).
 195 Some areas of low variance also show regions that retreat in all ensemble members, including the collapse of Pine Island Glacier and the Ross and Filchner-Ronne ice shelves.

For each parameter, we calculate the range in sea level contribution between ensemble members with minimal and maximal values of that parameter. This provides a rudimentary measure of sensitivity, which we can compare across parameters. Figure 5 highlights the timescale-dependance of this sensitivity, with different relative importance of parameters at 2100 and 2300.

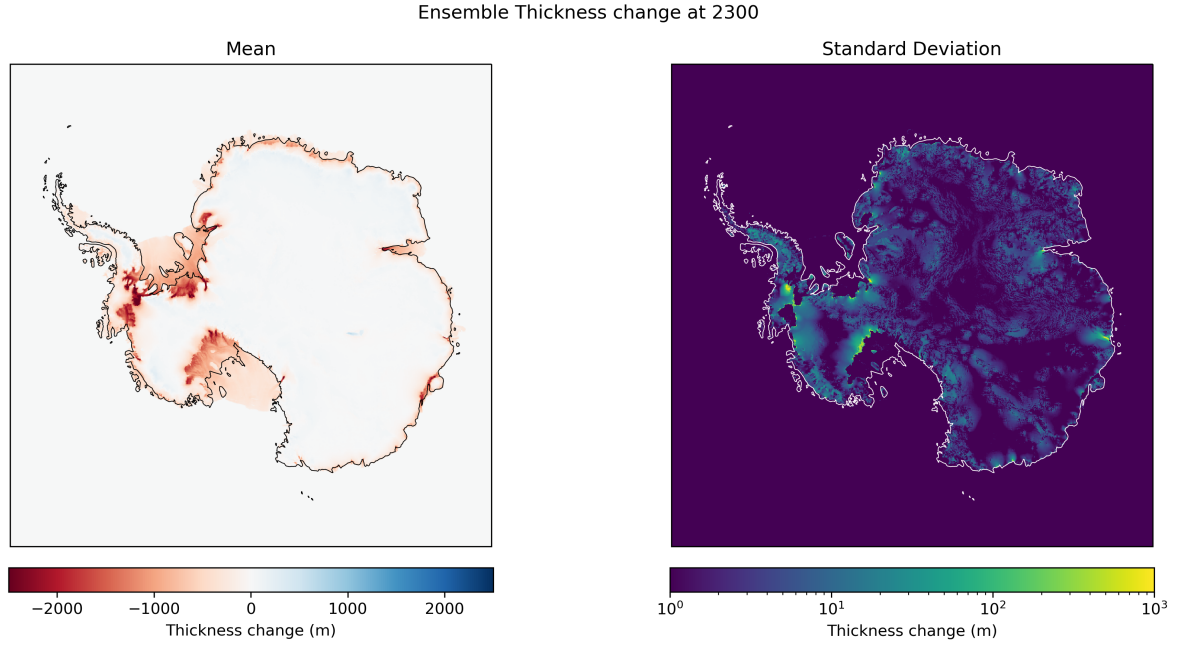


Figure 4. Timeseries of Antarctic sea level contribution for high and low values of each perturbed parameter.

At 2100, the largest sensitivity is to ice shelf basal melt, γ , but this reduces in relative terms by 2300 as other parameters become more important. Sensitivity to the flow law exponent, n , and the basal sliding exponent, m , both increase over time and are comparable to γ by 2300. Sensitivities to glacial isostatic adjustment (UMV) and bed friction regularisation (a_C) are negligible at 2100 and remain small by 2300. Sensitivity to regularisation of the viscosity coefficient, a_ϕ , is the second largest of all parameters by 2100 but becomes less significant relative to other parameters by 2300.

4 Discussion

The most direct comparison for our results is with (cite Rosier), who simulate the Amundsen Sea Embayment (ASE) up to 2100 using the $\tilde{U}a$ ice sheet model. Inversions in $\tilde{U}a$ are similar to those in BISICLES, using the adjoint method to solve for basal friction and the rate factor, which is analogous to the viscosity coefficient in BISICLES insofar as it controls ice rheology. Similar to Rosier et al., we find that regularisation parameters in the initialisation can be a significant source of uncertainty in century-scale sea level projections. However, there are some notable differences between our results.

Both studies agree that sensitivity to ‘rheology-related’ regularisation (rate factor in $\tilde{U}a$, viscosity coefficient in BISICLES) is higher than sensitivity to bed friction regularisation. In BISICLES, the viscosity coefficient plays a critical role in dynamics near to the present-day grounding line, which are highly influential in short-term sea level projections. However, we find that

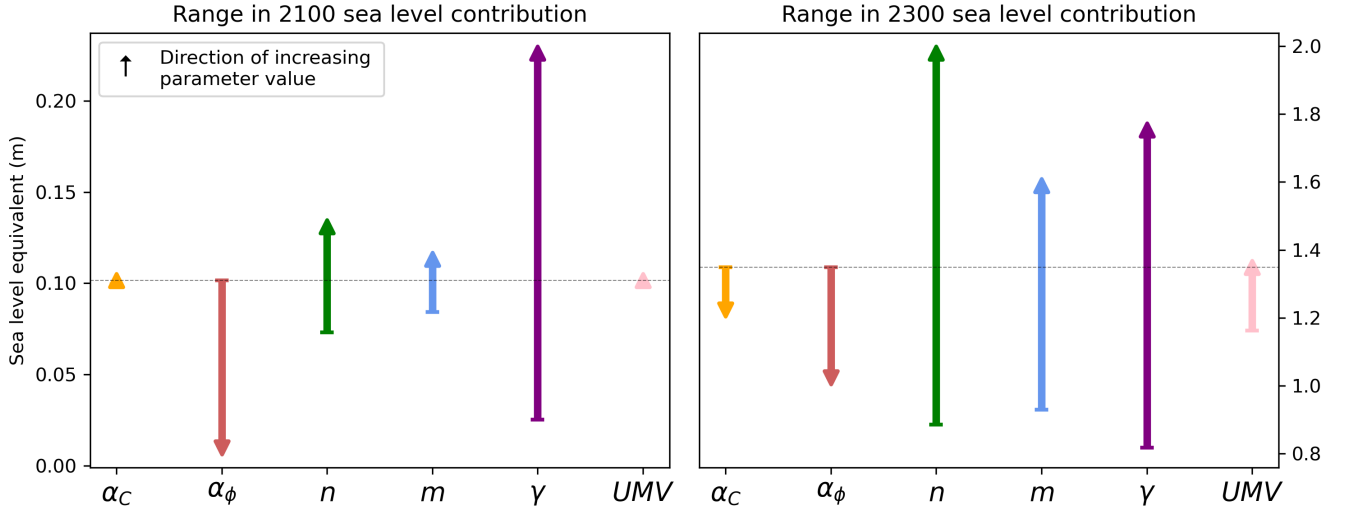


Figure 5. Sensitivity of Antarctic sea level contribution to each perturbed parameter at (a) 2100 and (b) 2300. Arrows denote the range in sea level contribution between ensemble members with minimal and maximal values of that parameter, pointing from minimal to maximal. Arrows pointing upwards/downwards therefore indicate a parameter that increases/decreases sea level contribution with higher values. The dashed horizontal line indicates the sea level contribution in a simulation with all parameters held at their default value.

this sensitivity decreases on longer timescales, possibly because the ice sheet retreats past present-day grounding lines into
 215 regions where the viscosity coefficient is mostly uniform.

Basal friction plays a role throughout the ice sheet and particularly in the ice streams, which may explain why sensitivity to its regularisation increases relative to rheology-related regularisation on longer timescales. However, sensitivity to bed friction regularisation in BISICLES is much lower than in $\dot{U}a$ when compared against other commonly perturbed parameters. This may be due to differences in the domain and resolution of the models. The inclusion of slow-moving parts of East Antarctica in our
 220 simulations introduces very high friction values that may dominate the Tikhonov penalty function, focusing the smoothing on areas of Antarctica that are less dynamically important to century-scale sea level projections. BISICLES is also known to be sensitive to grid resolution up to at least 500 m (cite steph 2016), and our 1 km resolution may be too coarse to fully capture the influence of bed friction regularisation on ice dynamics.

The timescale dependance of our results highlight the importance of considering the projection period and forcing when
 225 assessing sources of uncertainty. Because of the very high CESM2-WACCM forcing post-2100, the major ice shelves collapse in all ensemble members before 2300, which reduces sensitivity to ice shelf basal melt relative to other parameters on this timescale. Similarly, the collapse of Pine Island Glacier in all ensemble members affects the sensitivity of our outputs to key parameters in that region, which could include bed friction regularisation. If we had used a less extreme forcing — either a lower-emission scenario or a different climate model — we may have found different patterns of sensitivity. Rosier et al find

230 that parameters generally have higher sensitivity in a high-emissions scenario compared to a low-emissions scenario, but that this pattern is often reversed for initialisation parameters, supporting this idea.

5 Conclusions

We have explored the sensitivity of Antarctic sea level contribution to regularisation hyperparameters in the inverse problem and compared it with other commonly perturbed parameters. Using the BISICLES ice sheet model, the inverse problem solves
235 for a spatially-varying friction coefficient (C) and viscosity coefficient (ϕ). We find that sensitivity to friction-regularisation is much lower than in an equivalent study for the Amundsen Sea Embayment and suggest that this is due to a difference in model domains and resolution, particularly the inclusion of large slow-moving regions of East Antarctica in our experiment. We find that sensitivity to viscosity-regularisation is high relative to other parameters on a century-timescale, but that this sensitivity declines over time as dynamical parameters dominate the multi-century response. This remains a valid approach to exploring
240 initial state uncertainty in modelling experiments that initialise using inverse methods. However, we advise future experiments of this kind to consider the model, domain, resolution, and timescale of their experiment before choosing parameters to perturb, and warn that even if these parameters are included, it may not even then fully sample the uncertainty associated with the inverse problem.

Code availability. TEXT

245 *Data availability.* TEXT

Code and data availability. TEXT

Sample availability. TEXT

Video supplement. TEXT

Appendix A

250 **A1**

Author contributions. TEXT

Competing interests. TEXT

Disclaimer. TEXT

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